Layout-Preserving Translation of Scientific Documents

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Abstract

Researchers increasingly read papers written in a second language. Machine translation helps, but copying text out of a PDF destroys figures, tables and equations. We present a method that replaces only the running text of a paper while leaving every figure, table and formula exactly where it was. Our system detects the regions that must not change, translates the remaining paragraphs, and typesets the result into the original boxes.

1 Introduction

Reading a paper is a visual task. Readers move between a paragraph and the figure it describes, and they rely on the position of each element on the page. A translation that changes the layout forces the reader to search for content that used to be in a familiar place.

We control the trade-off between fidelity and readability with a single parameter α, which limits how much a paragraph may shrink to fit its box.

2 Method

Each text block is classified as body text, heading, caption or footnote. Blocks that belong to figures, tables or equations are left untouched. The remaining blocks are sent to the translation service, and the Japanese text is set with line-breaking rules so that punctuation never starts a line.

$$s = \min(1, \, w / w_0) \tag{1}$$

Here the scale factor is the ratio between the width of the box and the natural width of the translated text. Paragraphs that already fit are never enlarged.

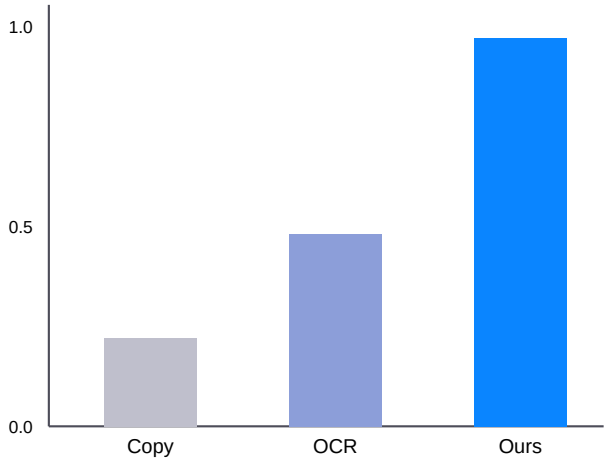

Figure 1: Layout similarity between the original and the translated paper.

3 Results

Figure 1 compares three approaches. Copying text into a translator keeps none of the layout, and optical character recognition loses most equations. Our method keeps the page structure intact while producing readable Japanese text.